

ASSIGNMENT BRIEF

HTU Course No: 00203420	HTU Course Name: Secure Coding
BTEC Unit Code:	BTEC UNIT Name:



Student Name/ID Number/Section	
HTU Course Number and Title	00203420 Secure Coding
BTEC Unit Code and Title	
Academic Year	2025-2026 2
Assignment Author	Safaa Hriez
Course Tutor	Safaa Hriez
Assignment Title	ShipTrack
Assignment Ref No	1
Issue Date	11/04/2026
Formative Assessment dates	From 11/04/2026 to 04/06/2026
Submission Date	07/06/2026
IV Name & Date	Qutaiba Albluwi 10/04/2026

Submission Format

The submission for this assignment is:

1. An individual written detailed report that answered the questions illustrated in the Assignment Brief and Guidance section.
2. An application that met the requirements presented in the Assignment Brief and Guidance section.

An oral discussion or its equivalent will be conducted after the submission of the report and the application. Instructions, date, and time for the discussion will be provided later. A witness statement or observation record is considered as evidence for this part.

Report guidelines:

In your report, you should make use of headings, paragraphs, and subsections as appropriate. The expected word limit is about 5000 word, although you will not be penalised for exceeding the total word limit. Do your best to be within the word limit. Your report should be:

1. In a form of a word file submitted via the elearning.
2. Written in a formal business style using Times New Roman and font size 12.

You are required to submit the declaration form to confirm that the submitted work is entirely your own and that you fully understand the consequences of plagiarism.

Unit Learning Outcomes

LO1 Identify the standards involved in the development of the efficient and secure code

LO2 Apply the quality and security patterns in the software development life cycle (SDLC)

LO3 Ability to apply security-driven design and implementation best practices in the software projects

LO4 Ability to test the quality and security of the software

Assignment Brief and Guidance

You are a software developer at **SkyRoute Solutions**, a leading company known for delivering secure, scalable, and high-performance software solutions for the logistics and delivery industry. SkyRoute has built a strong reputation for innovation, reliability, and customer-centric systems.

A package delivery company has approached SkyRoute Solutions to develop a new system that will streamline their delivery operations and improve efficiency. Recognizing your expertise, the company has assigned this project to you, trusting that you will deliver a system that aligns with their high standards of quality, security, and usability.

Application Name: ShipTrack

Application Description:

ShipTrack is an application designed to optimize package delivery operations by managing shipments, delivery personnel, and tracking records for both customers and staff. The following are some of the system requirements:

Requirements

1. Dispatcher:

- The dispatcher should be able to log in to the system.
- The dispatcher should be able to assign deliveries to drivers.
- The dispatcher should be able to view and update their personnel information.
- The dispatcher should be able to update delivery status ("pending", "in transit", "delivered").

2. Customer:

- The customer should be able to register in the application by providing name, ID number, and contact number.
- The customer should be able to log in to the system.
- The customer should be able to create shipment requests.
- The customer should be able to track the status of their packages.
- The customer should be able to view and update their personal information.

3. Delivery personnel:

- Delivery personnel should be able to log in to the system.
- They should be able to view assigned deliveries.
- They should be able to update delivery status ("picked up", "in transit", "delivered").

4. System Admin:

- The system admin should be able to register new dispatcher, including name, ID number, and contact number.
- The dispatcher should be able to register new delivery personnel, including name, ID number, and contact number.
- The system admin should be able to remove dispatcher or delivery personnel.
- The system admin should be able to define password strength requirements for all users, including:
 - Minimum number of characters
 - Minimum number of uppercase letters
 - Minimum number of lowercase letters
 - Minimum number of digits (0–9)
 - Minimum number of special characters
- The system admin should be able to define the maximum number of login attempts.
- The system admin should be able to lock and unlock user accounts

Important Notes:

- Ensure that the login feature is implemented securely using appropriate cryptographic standards to pass this course.
- Your code should use the code versioning methods to meet all the criteria.

You have to submit the code for the application and a report answering the following questions:

1. List two security principles that were developed in your application showing the code that present these principles. **Explain the benefits for each principle.**
2. List two quality principles that were illustrated in your application showing the code that present these principles. **Explain the benefits for each principle.**
3. Draw the use case diagram and the misuse case diagram for the application.
4. Create a data flow diagram.
5. Analyze the created data flow diagram for potential threats (Minimum 10 threats from each category).
6. Generate a full threat modeling report.
7. What types of testing methods can be used to evaluate whether users are satisfied with the application and whether it meets their requirements? What are the benefits of these methods?
8. What type of security testing methods will you use and what are their benefits?
9. Apply quality-based testing techniques such as unit testing by showing **the test plan** and providing screenshots for the testing.
10. Apply security-based testing techniques such as fuzz testing and provide screenshots for the testing.
11. Apply code review analysis and provide screenshots.

Learning Outcomes and Assessment Criteria			
Learning Outcome	Pass	Merit	Distinction
LO1 Identify the standards involved in the development of the efficient and secure code	P1 Discuss the principles and concepts of quality software. P2 Discuss the principles and concepts of secure software	M1 Analyse the benefits of each principle and how to use it during the development of the software	D1 Determine the mitigation of the potential threats and generate a full report for the threat model.
	P3 Design quality and security software using industry patterns. P4 Create a data flow diagram model.	M2 Analyze the data flow diagram for potential threats.	
LO3 Ability to apply security-driven design and implementation best practices in the software projects	P5 Develop a specific feature in a software project complying with the principles of the quality and security. P6 Define and use cryptography standards.	M3 Develop quality-driven design for a specific software project.	D2 Develop security-driven design for a specific software project D3 Apply code versioning
LO4 Ability to test the quality and security of the software	P7 Understand the basic concepts and definitions of testing. P8 Discuss the security testing methods.	M4 Apply quality-based testing techniques.	D4 Apply security-based testing techniques. D5 Apply code review analysis.